Abylay Bissekenov

Astrophysicist

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Education

PhD **Xi'an Jiaotong-Liverpool University**, Physics — Suzhou, China
Mar 2024 – Dec 2026

MS **Nazarbayev University**, Physics — Astana, Kazakhstan
Aug 2021 – June 2023
- GPA: 3.5/4.00
- Graduated with honors

BS **Aktobe Regional State University**, Physics — Aktobe, Kazakhstan
Aug 2021 – June 2023
- GPA: 3.64/4.00
- Graduated with honors

Experience

Xi'an Jiaotong-Liverpool University, Research Assistant — Suzhou, China
Mar 2024 – present
2 years 4 months

Xi'an Jiaotong-Liverpool University, Teaching Assistant — Suzhou, China
Mar 2024 – June 2026
2 years 4 months
- Over 500 hours of TAship experience

Energetic Cosmos Laboratory, Nazarbayev University, Research Assistant — Astana, Kazakhstan
June 2022 – Dec 2024
2 years 7 months

Projects

FlashInfer — Jan 2023 – present
Open-source library for high-performance LLM inference kernels
- Achieved 2.8x speedup over baseline attention implementations on A100 GPUs
- Adopted by 3 major AI labs, 8,500+ GitHub stars, 200+ contributors

NeuralPrune — Jan 2021
Automated neural network pruning toolkit with differentiable masks
- Reduced model size by 90% with less than 1% accuracy degradation on ImageNet
- Featured in PyTorch ecosystem tools, 4,200+ GitHub stars

Publications

Sparse Mixture-of-Experts at Scale: Efficient Routing for Trillion-Parameter Models — July 2023
John Doe, Sarah Williams, David Park
10.1234/neurips.2023.1234 (NeurIPS 2023)

Neural Architecture Search via Differentiable Pruning Dec 2022
James Liu, *John Doe*
10.1234/neurips.2022.5678 (NeurIPS 2022, Spotlight)

Multi-Agent Reinforcement Learning with Emergent Communication July 2022
Maria Garcia, *John Doe*, Tom Anderson
10.1234/icml.2022.9012 (ICML 2022)

On-Device Model Compression via Learned Quantization May 2021
John Doe, Kevin Wu
10.1234/iclr.2021.3456 (ICLR 2021, Best Paper Award)

Selected Honors

- MIT Technology Review 35 Under 35 Innovators (2024)
- Forbes 30 Under 30 in Enterprise Technology (2024)
- ACM Doctoral Dissertation Award Honorable Mention (2023)
- Google PhD Fellowship in Machine Learning (2020 – 2023)
- Fulbright Scholarship for Graduate Studies (2018)

Skills

Languages: Python, C++, CUDA, Rust, Julia
ML Frameworks: PyTorch, JAX, TensorFlow, Triton, ONNX
Infrastructure: Kubernetes, Ray, distributed training, AWS, GCP
Research Areas: Neural architecture search, model compression, efficient inference, multi-agent RL

Patents

1. Adaptive Quantization for Neural Network Inference on Edge Devices (US Patent 11,234,567)
2. Dynamic Sparsity Patterns for Efficient Transformer Attention (US Patent 11,345,678)
3. Hardware-Aware Neural Architecture Search Method (US Patent 11,456,789)

Invited Talks

4. Scaling Laws for Efficient Inference — Stanford HAI Symposium (2024)
3. Building AI Infrastructure for the Next Decade — TechCrunch Disrupt (2024)
2. From Research to Production: Lessons in ML Systems — NeurIPS Workshop (2023)
1. Efficient Deep Learning: A Practitioner's Perspective — Google Tech Talk (2022)